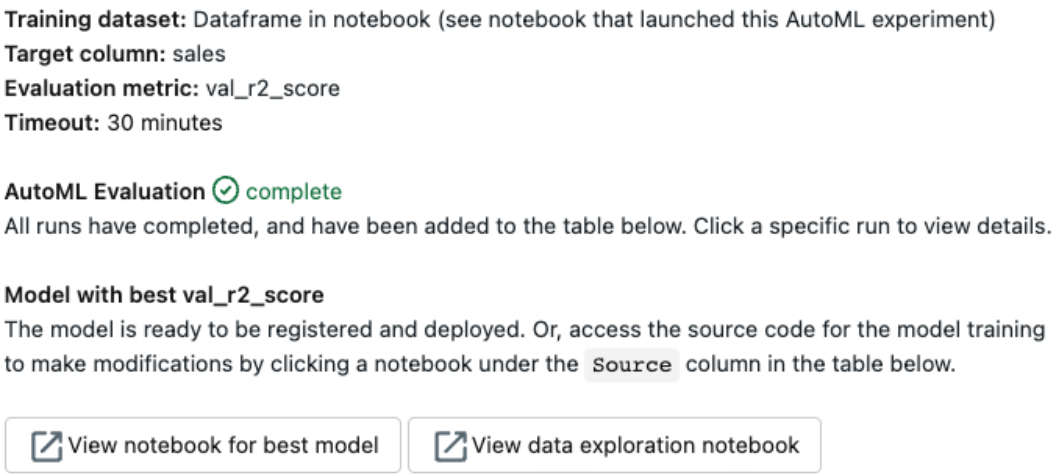


Figure 4.12 – The AutoML experiment page with the links to view the notebook for the best model and the data exploration notebook

The exploration notebook uses `ydata-profiling`, formerly referred to as pandas' profiler library, to generate statistics and summarize data for all the fields in the table. It also provides alerts on fields with high correlation issues, which could negatively impact models. These warnings are also available in the MLflow experiment UI, as shown in *Figure 4.13*.

Severity	Type	Affected Data	Action
Medium	More than 5% of values are null.	local_holiday_type national_holiday_type regional_holiday_type	AutoML imputed the null values.
Low	Data exploration notebook ran on a subset of rows because dataset is too big		Modify the data exploration notebook and rerun it to profile the entire dataset.
Low	High correlation columns	onpromotion national_holiday_type date family lag10_oil_price +4 more	Correlations found. Refer to data exploration notebook for more details.

Figure 4.13 – AutoML warnings called out during the experiment run

Look through the data exploration notebook for an overview of your data, from summary statistics to thorough profiles for each variable.

We have now explored two tools for data exploration: Databricks Assistant and the `ydata-profiling` library. These are great places to start for many classical ML projects. Next, we'll discuss a more advanced data format and how you can use the DI Platform to explore it: data embeddings and vector search. Embeddings are advanced transformations that translate complex, unstructured data into a numerical format conducive to ML algorithms, capturing intricate relationships within the data that are pivotal for sophisticated models.